

Each question is 25 points. This exam is from 9:30AM to 10:45AM

Good luck ☺

1. How many finite sets of integers are there? Prove your answer.There are exactly $\aleph_0$ (countably infinite) finite sets of integers.

Proof:

1) There are at least $\aleph_0$ such sets because: $\{1\}, \{2\}, \{3\}, \dots$ are all different finite sets.

Since there are infinitely many integers, there are infinitely many sets.

Now, we show there are at most $\aleph_0$ finite sets.

Every finite set of integers can be written as a finite list of integers in increasing order.

Since $\mathbb{Z}$ is countable, finite tuples of integers are countable, a countable union of countable sets is countable, the set of all finite subsets of $\mathbb{Z}$ is countable. $\therefore$ the collection of finite sets is exactly $\aleph_0$.

• Note that $\aleph_0$ is the cardinality of the set of natural numbers.

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2. Is the set of "fuzzy" total functions from $\mathbb{N}$ to $\{\overset{0}{\text{Yes}}, \overset{1}{\text{No}}, \overset{2}{\text{Perhaps}}\}$ countable? Prove it.

No, it's uncountable
Proof: by Cantor's Diagonalization

Identify:

Yes = 0

No = 1

Maybe = 2

Assume for contradiction that all such total functions can be listed. :-

$b_0, b_1, b_2, \dots$

Define a new function: $g(n) = (b_n(n) + 1) \bmod 3$.

Then for every n : $g(n) \neq b_n(n)$.

Thus g differs from every function in the list.

This contradicts the assumption that all total functions were listed.

Therefore, the set of "fuzzy" total functions from $\mathbb{N}$ to $\{\text{Yes}, \text{No}, \text{Perhaps}\}$ is uncountable.

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3. Is $K = \{i \mid M_i(i) \text{ halts}\}$ an r.e. set? Is it recursive? Is the complement of K r.e.? Prove it.

a) Is K r.e.?

Yes, on input i :

- Simulate $M_i(i)$.
- If halts, accept.
- If it doesn't halt, run forever.

Thus, the machine accepts the elements of K . Therefore, K is r.e.

b) Is K recursive?

No, assume K is recursive. Then there exists a machine D that decides whether $M_i(i)$ halts.

Construct a machine E that:

- If $D(i)$ "halts", then loop forever.
- If $D(i)$ says does not halt, then halt.

Let e be the index of E .

Now consider $E(e)$:

If $D(e)$ says it halts,
→ $E(e)$ loops

If $D(e)$ says it loops,
→ $E(e)$ halts

Contradiction. Therefore K is not recursive.

c) Is $\bar{K}$ r.e.?

No, $\bar{K}$ is not r.e.

$\bar{K} = \{i \mid M_i(i) \text{ does not halt}\}$.

Proof:

From a) proven that K is r.e.

→ Simulate $M_i(i)$ and accept if it halts.

But, since we know that K is not decidable (otherwise we could have proven by running both recognizers in parallel).

In other words, if $\bar{K}$ were r.e., K would be recursive. From b) we know this is not true.

This is a contradiction.

Therefore, $\bar{K}$ is not r.e.

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4. Is $R = \{j \text{ such that } M_j \text{ accepts at least } j \text{ inputs}\}$ an r.e. set? Is it recursive? Prove both.

a) R is an r.e. set.

Proof: Construct a Turing Machine T that on input:

- 1) Dovetail the simulations of $M_j(x)$ for all inputs $x=0,1,2,\dots$
- 2) Keep a counter of how many distinct inputs are observed to be accepted.
- 3) If the counter reaches j , accept.
 - If $j \in R$, then M_j accepts at least j inputs. Each accepting computation will eventually be discovered during dovetailing, so the counter will reach j and T halts and accepts.
 - If $j \notin R$, then fewer than j inputs are accepted, so the counter never reaches j , and T runs forever.

Thus T recognizes R , so R is r.e.

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b) R is not recursive.

Proof: (reduction from K).

Let $K = \{e \mid M_e(e) \text{ halts}\}$ which is not recursive.

We reduce K to R .

Given e , construct a machine N that behaves as follows:

- On input x :
- if $x < k$, simulate $M_e(e)$.
 - if $M_e(e)$ halts, accept.
 - Otherwise, loop.
- if $x \geq k$, reject.

By the reduction theorem, there exists an index k such that $M_k = N$.

Now:

if $M_e(e)$ halts, then M_k accepts all inputs $0, 1, \dots, k-1$,

so it accepts at least k inputs.

• $k \in R$: If $M_e(e)$ does not halt, then M_k accepts no inputs.

• $k \notin R$: Thus: $e \in K \iff k \in R$

Therefore, $K \leq_m R$

Since K is not recursive R cannot be recursive.

$\therefore R$ is not recursive.